DiscoGMPI Numerical Implementation Notes

DiscoGMPI project

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1 Polynomial Approximation Space

Each scalar component of $\boldsymbol{E}$ and $\boldsymbol{H}$ is approximated independently on every tetrahedron by a polynomial in

$$\mathbb{P}^N(K) = \{p(x, y, z) : \deg p \leq N\}. \tag{1}$$

The number of volume nodes is

$$N_p = \frac{(N+1)(N+2)(N+3)}{6}, \tag{2}$$

and the number of nodes on a triangular face is

$$N_{fp} = \frac{(N+1)(N+2)}{2}. \tag{3}$$

The Maxwell state stores six arrays of size $N_p \times K$, where K is the number of rank-local tetrahedra including ghosts in distributed runs.

2 Reference Tetrahedron and Nodal Sets

The reference tetrahedron has vertices

$$(-1,-1,-1), \quad (1,-1,-1), \quad (-1,1,-1), \quad (-1,-1,1). \tag{4}$$

`build_reference_tet(N)` constructs equispaced nodal points on this reference element by default. The optional call `build_reference_tet(N; nodal_set=:warp_blend)` constructs the Hesthaven–Warburton tetrahedral warp-blend nodes. For non-equispaced tetrahedral nodes, the DG face operators build each triangular face mass matrix from the actual face-node coordinates, so the trace space remains consistent with the volume nodal set. High-order ParaView output currently assumes VTK Lagrange/equispaced node ordering; warp-blend runs therefore require sampled visualization output or an output-specific interpolation path before the high-order Lagrange `.vtu` files can be interpreted directly.

An orthonormal modal tetrahedral basis is evaluated at the nodes to form the Vandermonde matrix

$$V_{ij} = \phi_j(r_i, s_i, t_i). \tag{5}$$

The inverse Vandermonde maps nodal values to modal coefficients and represents nodal basis functions as modal combinations.

3 Quadrature and Cubature

Independent diagnostic integration uses Jaskowiec–Sukumar tetrahedral cubature rules through `get_JaskowiecSukumar_cubature(order)`. Production diagnostics commonly use

$$q = \max(2, 2N+4), \tag{6}$$

so field errors, energy-density errors, chirality, and momentum diagnostics are not dominated by low-order quadrature.

4 Mass Matrices

On the reference tetrahedron, the nodal mass matrix is

$$M = V^{-T} V^{-1}, \tag{7}$$

because the modal basis is orthonormal. For an affine physical element,

$$M_K = |J_K| M. \tag{8}$$

The DG energy diagnostic uses this mass matrix directly:

$$\mathcal{H}_h = \frac{1}{2} \sum_K |J_K| \left[\varepsilon \sum_a E_{a,K}^T M E_{a,K} + \mu \sum_a H_{a,K}^T M H_{a,K} \right]. \tag{9}$$

5 Derivative and Stiffness Matrices

Reference derivative matrices are formed from modal derivative Vandermonde matrices:

$$D_r = V_r V^{-1}, \qquad D_s = V_s V^{-1}, \qquad D_t = V_t V^{-1}. \tag{10}$$

The weak stiffness matrices are

$$S_r = M D_r, \qquad S_s = M D_s, \qquad S_t = M D_t. \tag{11}$$

Metric terms transform these into physical derivative matrices D_x, D_y, D_z and weak operators S_x, S_y, S_z. The Hesthaven–Warburton path uses strong derivative operators for volume curls. The Poisson-bracket path uses weak operators for one partition and transposed weak operators for the complementary partition.

6 Face Operators and Connectivity

Reference face operators identify the N_{fp} nodes on each triangular face and build face mass/lifting operators. A face residual contributes conceptually as

$$M_K^{-1} M_{f,K} f \tag{12}$$

to the nodal right-hand side.

DG topology is constructed by sorting the three vertex IDs of each tetrahedral face. A face appearing once is a physical boundary face, a face appearing twice is an interior face, and a face appearing more than twice is rejected as non-manifold. Interior trace maps store minus and plus element IDs, local face IDs, face-node lists, plus-to-minus nodal permutations, outward normals, and face areas.

7 Distributed Ownership and Periodic Maps

In MPI runs, each rank owns a subset of global tetrahedra and stores a one-face halo. Volume work and diagnostics use owned elements only. Ghost fields are updated by exchanging face traces before surface flux evaluation.

Periodic boundaries are directed face pairs. The pairing logic matches boundary face centroids after a translation and constructs node permutations to align plus-side and minus-side nodes. For the default rectangular periodic box, the boundary tags are

$$\begin{aligned} 1 &\leftrightarrow 2 \quad \text{x-min / x-max} \\ 3 &\leftrightarrow 4 \quad \text{y-min / y-max} \\ 5 &\leftrightarrow 6 \quad \text{z-min / z-max.} \end{aligned}$$

Periodic tags should be handled by the periodic exchange object and should remain unregistered as physical boundary conditions in the ordinary registry.

8 Metric Terms and Mesh Quality

For each affine tetrahedron, the Jacobian of the map from reference coordinates to physical coordinates gives

$$\nabla_x u = J_K^{-T} \nabla_{r,s,t} u. \tag{13}$$

The determinant $|J_K|$ scales volume integrals, and face geometry supplies normals and areas.

The convergence CSV records mesh-quality metrics: volume min/max/total/ratio, edge min/max/ratio, element-size min/max/ratio based on maximum edge length, and tetrahedron mean-ratio quality. The mean-ratio quality is

$$q_K = \frac{12(3V_K)^{2/3}}{\sum_{e\in K}|e|^2}, \tag{14}$$

where the denominator sums the squared lengths of the six tetrahedron edges.

9 Flux Evaluation

Flux evaluation proceeds by gathering minus and plus traces, aligning plus-side nodes to the minus-side ordering, evaluating the face correction, and lifting it into the volume. The centered Hesthaven–Warburton correction is

$$F_E^c = \frac{1}{2}\boldsymbol{n}\times(\boldsymbol{H}^+ - \boldsymbol{H}^-), \qquad F_H^c = -\frac{1}{2}\boldsymbol{n}\times(\boldsymbol{E}^+ - \boldsymbol{E}^-). \tag{15}$$

The upwind correction adds impedance-weighted tangential penalties:

$$-\frac{Y}{2}\boldsymbol{n}\times(\boldsymbol{n}\times(\boldsymbol{E}^+ - \boldsymbol{E}^-)), \tag{16}$$

$$-\frac{Z}{2}\boldsymbol{n}\times(\boldsymbol{n}\times(\boldsymbol{H}^+ - \boldsymbol{H}^-)). \tag{17}$$

The Poisson-bracket operator uses

$$F_E^{PB} = -\frac{1}{2\varepsilon}\boldsymbol{n}\times(\boldsymbol{H}^- - \boldsymbol{H}^+), \qquad F_H^{PB} = -\frac{1}{2\mu}\boldsymbol{n}\times(\boldsymbol{E}^- + \boldsymbol{E}^+). \tag{18}$$

The implemented Poisson-bracket surface operator supports the centered and alternating variants. Upwind fluxes are rejected for this formulation because they are dissipative and do not define the partitioned Poisson-bracket operator used by the ESPRK path.

10 Magnetic-Charge Null-Invariant Diagnostic

For this formulation, the magnetic-field weak equation has the bracket-induced curl form

$$\left(\mu\dot{\boldsymbol{H}}_h, \boldsymbol{w}\right)_{\mathcal{T}_h} = -\left(\boldsymbol{E}_h, \nabla\times\boldsymbol{w}\right)_{\mathcal{T}_h} + \left\langle \widehat{\boldsymbol{E}}_h, \boldsymbol{n}\times\boldsymbol{w}\right\rangle_{\partial\mathcal{T}_h}. \tag{19}$$

In algebraic form this can be written schematically as

$$M_H\dot{\boldsymbol{h}} = C_{HE}\boldsymbol{e}, \tag{20}$$

where C_{HE} denotes the actual matrix-free operator assembled by the code: volume curl terms, material interior faces, periodic faces, MPI halo faces, and physical boundary traces.

The signed magnetic-charge diagnostic currently monitored by `src/MaxwellDiagnostics.jl` is the strong-divergence integral

$$Q_{H,h} = \sum_{K\in\mathcal{T}_h^{owned}} \mu_K|J_K|\,\mathbf{1}^T M\left(D_xH_{x,K} + D_yH_{y,K} + D_zH_{z,K}\right), \tag{21}$$

followed by an MPI sum over ranks. The quadrature diagnostics report the same quantity in signed form, its absolute-value analogue, and the local broken divergence norms. The latter are useful local diagnostics, but they are not by themselves the algebraic invariant of the implemented bracket-induced curl operator.

If the monitored charge is written as $Q_{H,h} = \boldsymbol{q}_H^T M_H \boldsymbol{h}$, the semidiscrete conservation condition is

$$\frac{\mathrm{d}Q_{H,h}}{\mathrm{d}t} = \boldsymbol{q}_H^T C_{HE} \boldsymbol{e} = 0 \qquad \text{for every } \boldsymbol{e}, \tag{22}$$

or, equivalently, $\boldsymbol{q}_H^T C_{HE} = 0$. If the same diagnostic is represented through a discrete strong-divergence matrix D_H, the corresponding null-space test is $D_H M_H^{-1} C_{HE} = 0$.

The diagnostic routine in `src/MaxwellDiagnostics.jl` evaluates this condition with the production matrix-free RHS rather than an independently defined magnetic normal flux. It compares the full periodic RHS derivative with the sum of the same split RHS pieces:

```
distributed_magnetic_charge_rhs_breakdown
maxwell_rhs_periodic!
maxwell_volume_rhs!
add_material_interior_surface_rhs!
add_distributed_periodic_surface_rhs!
add_material_boundary_surface_rhs!
```

The focused MPI test `test/test_mpi_magnetic_charge_null_invariant.jl` applies representative electric-field coefficient vectors on a structured periodic box and checks the null residual for centered and alternating Poisson-bracket fluxes on 1, 2, and 4 MPI ranks. The validation orchestrator `examples/validate_distributed_maxwell_matrix.jl` includes this test in the MPI release-gate checks.

The observed residuals are at roundoff level for the tested structured periodic cases. Therefore a very small long-time drift of the signed global magnetic charge is consistent with accumulated floating-point error, provided this algebraic null-invariant test remains satisfied. A nonzero broken elementwise divergence norm should be interpreted separately, since volume and skeleton terms can cancel in the bracket-induced invariant.

11 Boundary Trace Construction

PEC and PMC boundary conditions are handled by reflected exterior traces:

$$\boldsymbol{E}^+_{PEC} = -\boldsymbol{E}^- + 2(\boldsymbol{n} \cdot \boldsymbol{E}^-)\boldsymbol{n}, \qquad \boldsymbol{H}^+_{PEC} = \boldsymbol{H}^-, \tag{23}$$

$$\boldsymbol{E}^+_{PMC} = \boldsymbol{E}^-, \qquad \boldsymbol{H}^+_{PMC} = -\boldsymbol{H}^- + 2(\boldsymbol{n} \cdot \boldsymbol{H}^-)\boldsymbol{n}. \tag{24}$$

Periodic boundaries use true traces from paired periodic faces rather than reflected traces.

12 Time Integration

DiscoGMPI contains conventional explicit Runge–Kutta methods and explicit partitioned symplectic Runge–Kutta methods. For a semidiscrete ODE $\dot{U} = R(U)$, explicit RK methods use

$$U^{(i)} = U^n + \Delta t \sum_{j<i} a_{ij} K_j, \qquad K_i = R(U^{(i)}), \qquad U^{n+1} = U^n + \Delta t \sum_i b_i K_i. \tag{25}$$

For Poisson-bracket Maxwell runs, the recommended method is ESPRK with magnetic-first partitioning. The convergence drivers usually set ESPRK order to $N+1$ for DG order N so time error does not dominate the spatial study.

13 Time-Domain PEC Sphere RCS

The metallic-sphere RCS workflow is implemented by the following source files:

- `examples/distributed_metallic_sphere_scattering.jl`
  Production MPI scattering driver.
- `src/MaxwellExactSolutions.jl`
  Exact incident plane-wave fields.
- `src/solver/kernels/maxwell.jl`
  Incident-aware PEC boundary data.
- `examples/postprocess_metallic_sphere_mie_rcs.py`
  Mie-series comparison and plotting.
- `examples/validate_pec_sphere_boundary_residual.jl`
  PEC boundary-residual validation gate.
- `examples/validate_scattered_ex_hy_fields.jl`
  Qualitative ParaView scattered-field validation gate.
- `examples/validate_coarse_sphere_rcs.jl`
  Coarse RCS/Mie comparison validation gate.

The production scattering driver evolves the scattered field

$$\boldsymbol{E}_h = \boldsymbol{E}_{\mathrm{scat},h}, \qquad \boldsymbol{H}_h = \boldsymbol{H}_{\mathrm{scat},h},$$

not the total field. The initial scattered field is zero. The total field used for diagnostics and output is reconstructed as

$$\boldsymbol{E}_{\mathrm{tot},h} = \boldsymbol{E}_{\mathrm{scat},h} + \boldsymbol{E}_{\mathrm{inc}}, \qquad \boldsymbol{H}_{\mathrm{tot},h} = \boldsymbol{H}_{\mathrm{scat},h} + \boldsymbol{H}_{\mathrm{inc}}. \tag{26}$$

ParaView snapshots write `E_scat`, `H_scat`, `E_inc`, `H_inc`, `E_total`, `H_total`, and the scattered-field magnitudes. No incident-wave volume source is added in this driver; the incident field enters through the inhomogeneous PEC trace on the scatterer and through diagnostic reconstruction of the total field.

The incident plane wave is parameterized by a unit propagation direction $\widehat{\boldsymbol{d}}$, a unit polarization $\widehat{\boldsymbol{p}}$ with $\widehat{\boldsymbol{d}} \cdot \widehat{\boldsymbol{p}} = 0$, wavelength λ, electric amplitude A, and material constants ε, μ. The implemented sphere driver uses

$$\widehat{\boldsymbol{d}} = (0, 0, 1), \qquad \widehat{\boldsymbol{p}} = (1, 0, 0),$$

unless the code is changed. With

$$k = \frac{2\pi}{\lambda}, \qquad c = \frac{1}{\sqrt{\varepsilon\mu}}, \qquad \omega = ck, \qquad Z = \sqrt{\frac{\mu}{\varepsilon}},$$

the exact incident fields used by the driver are

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{x}, t) = A\,\widehat{\boldsymbol{p}}\,\cos\Big(k\widehat{\boldsymbol{d}} \cdot \boldsymbol{x} - \omega t + \phi_0\Big)\,, \tag{27}$$

$$\boldsymbol{H}_{\mathrm{inc}}(\boldsymbol{x}, t) = \frac{A}{Z}\,\Big(\widehat{\boldsymbol{d}} \times \widehat{\boldsymbol{p}}\Big)\cos\Big(k\widehat{\boldsymbol{d}} \cdot \boldsymbol{x} - \omega t + \phi_0\Big)\,. \tag{28}$$

The command-line option `-frequency` overrides the wavelength through $\lambda = c/f$. Otherwise $f = c/\lambda$.

The metallic surface is tagged as PEC; the default sphere mesh uses boundary tag 10, while tags $1, \ldots, 6$ are the outer Cartesian PML boundaries. For the scattered-field unknowns the PEC condition is imposed as

$$\boldsymbol{n} \times \boldsymbol{E}_{\mathrm{tot}} = 0 \quad \Longleftrightarrow \quad \boldsymbol{n} \times \boldsymbol{E}_{\mathrm{scat}} = -\boldsymbol{n} \times \boldsymbol{E}_{\mathrm{inc}}. \tag{29}$$

At a PEC face node the exterior electric trace is the homogeneous PEC reflection corrected by the incident tangential field:

$$\boldsymbol{E}^{+}_{\mathrm{scat}} = -\boldsymbol{E}^{-}_{\mathrm{scat}} + 2(\boldsymbol{n}\cdot\boldsymbol{E}^{-}_{\mathrm{scat}})\boldsymbol{n} - 2\left[\boldsymbol{E}_{\mathrm{inc}} - (\boldsymbol{n}\cdot\boldsymbol{E}_{\mathrm{inc}})\boldsymbol{n}\right], \qquad \boldsymbol{H}^{+}_{\mathrm{scat}} = \boldsymbol{H}^{-}_{\mathrm{scat}}. \tag{30}$$

This is implemented by `pec_incident_boundary_plus_trace!`. The reported PEC residual is

$$\|\boldsymbol{n}\times\boldsymbol{E}_{\mathrm{tot},h}\|_{L^2(\Gamma_{\mathrm{PEC}})} \tag{31}$$

and is written to `diagnostics/pec_boundary_residual.csv`.

The scattering problem is nonautonomous because the boundary trace depends on the incident field time. The ESPRK scheme used by the Poisson-bracket driver does not store classical Runge–Kutta abscissas c_i. It stores two composition-weight vectors, a_i for the first Maxwell partition and b_i for the second. For the H-first split used by the RCS driver, the first partition is H and the second is E. The driver reconstructs the physical substep clocks

$$t_i^H = t^n + \Delta t\sum_{j<i} a_j, \qquad t_i^E = t^n + \Delta t\sum_{j<i} b_j. \tag{32}$$

For composition substep i, the driver sets the boundary-data time to t_i^H before the H-partition right-hand side and to t_i^E before the E-partition right-hand side. After the full step it is set to t^{n+1}. This synchronization is performed in `timed_partitioned_symplectic_rk_step!`; using a single classical stage time would not match the active partition clock of this composition.

When RCS extraction is enabled, owned PEC surface faces are sampled on each MPI rank. The code forms the PEC surface current from the reconstructed total magnetic field,

$$\boldsymbol{J}_s(\boldsymbol{x},t) = \boldsymbol{n}\times\boldsymbol{H}_{\mathrm{tot},h}(\boldsymbol{x},t), \tag{33}$$

where the diagnostic normal stored for inner scatterer faces is the negated DG boundary normal. For each observation direction $\widehat{\boldsymbol{r}}$, with spherical basis $(\widehat{\boldsymbol{e}}_\theta,\widehat{\boldsymbol{e}}_\phi)$, the rank-local surface integral is accumulated with outgoing phase convention

$$\boldsymbol{I}(\widehat{\boldsymbol{r}},t) = \int_{\Gamma_{\mathrm{PEC}}} \boldsymbol{J}_s(\boldsymbol{x},t)\,\exp(-\mathrm{i}k\widehat{\boldsymbol{r}}\cdot\boldsymbol{x})\;\mathrm{d}S. \tag{34}$$

Only samples in the RCS time window contribute. The temporal window is the Hann function

$$w(t) = \frac{1}{2}\left[1-\cos\left(2\pi\frac{t-t_0}{t_1-t_0}\right)\right], \qquad t_0\le t\le t_1, \tag{35}$$

with zero contribution outside the interval. The accumulated complex phasor is formed as

$$\widehat{\boldsymbol{I}}(\widehat{\boldsymbol{r}}) = \frac{2}{\sum_m w(t_m)}\sum_m w(t_m)\,\exp(\mathrm{i}\omega t_m)\,\boldsymbol{I}(\widehat{\boldsymbol{r}},t_m). \tag{36}$$

The factor $2/\sum_m w(t_m)$ is the implemented real-cosine phasor normalization. Rank-local complex accumulators are summed with MPI reductions before the final RCS table is written.

The far-field amplitude used by `write_rcs_results` is

$$\boldsymbol{E}^{\infty}(\widehat{\boldsymbol{r}}) = \frac{\mathrm{i}kZ}{4\pi}\left[\widehat{\boldsymbol{r}}\left(\widehat{\boldsymbol{r}}\cdot\widehat{\boldsymbol{I}}\right) - \widehat{\boldsymbol{I}}\right]. \tag{37}$$

The code reports dimensional angular RCS values

$$\sigma(\theta,\phi) = 4\pi\frac{|E^{\infty}_{\theta}|^2 + |E^{\infty}_{\phi}|^2}{A^2}, \qquad E^{\infty}_{\theta} = \widehat{\boldsymbol{e}}_\theta\cdot\boldsymbol{E}^{\infty}, \quad E^{\infty}_{\phi} = \widehat{\boldsymbol{e}}_\phi\cdot\boldsymbol{E}^{\infty}. \tag{38}$$

The corresponding decibel column is $10 \log_{10} \sigma$. The driver writes angular patterns over the requested (θ, ϕ) grid to `diagnostics/rcs.csv`; it does not itself normalize by πa^2. The incident direction is fixed, and the requested observation directions are sampled independently, so the table is a bistatic angular pattern rather than a separate monostatic backscatter mode. In the default convention $\theta = 0$ points in the incident $+z$ direction and $\theta = 180^\circ$ is the backscatter direction. Normalization by the geometrical cross section πa^2, Mie-series comparison, LaTeX tables, and Cartesian/polar plots are performed by `examples/postprocess_metallic_sphere_mie_rcs.py` and by the coarse validation gate.

The mesh generator `examples/meshes/metallic_sphere_scattering.geo` creates a box-minus-sphere tetrahedral mesh with PEC sphere tag 10, outer PML tags $1, \ldots, 6$, and volume tag 7. The run directory records the resolved configuration, mesh path, generated balanced spatial partition, MPI rank count, time step, RCS window, and diagnostics cadence. The scattering driver currently partitions the root-loaded VTK mesh with a balanced spatial partition, builds the distributed DG discretization, computes owned-element volume work and owned PEC-face RCS contributions, and uses MPI reductions for global diagnostics and RCS accumulation.

14 CFL Condition

The basic timestep estimate is

$$\Delta t = \mathrm{CFL} \frac{\min_K h_K}{(2N+1)c}, \tag{39}$$

where c is the material wave speed. The CFL element size used by the timestep estimate is

$$h_K = \frac{3V_K}{A_K}, \tag{40}$$

with V_K the tetrahedron volume and A_K its total surface area. The final timestep is adjusted so the run lands exactly on the requested final time.

15 Mesh-Size Definitions

The code and CSV files distinguish several mesh-size measures:

- `target_h`: requested Gmsh size, typically L_x/N_x;
- `characteristic_h`: measured volume scale, typically $(|\Omega|/N_{el})^{1/3}$;
- `h_min`, `h_max`: quality sizes based on maximum tetrahedron edge length;
- CFL h_K: $3V_K/A_K$.

Production convergence rates use the measured characteristic mesh size so structured and independently generated unstructured families are treated consistently.

16 Structured and Unstructured Families

The cavity convergence driver supports structured unit-cube meshes and unstructured unit-cube Gmsh families. The periodic convergence driver supports structured $[0, 2] \times [0, 1] \times [0, 1]$ periodic box meshes and unstructured periodic Gmsh families. The CSV stores `mesh_family` so post-processing can separate structured and unstructured studies.

17 Expected Convergence Rates

For polynomial degree N, the current regression targets are aggregate electric-field rate $N + 1$ and aggregate magnetic-field rate N. Component errors and rates are recorded for

$$E_x,\ E_y,\ E_z,\ H_x,\ H_y,\ H_z. \tag{41}$$

Component rates are suppressed when the coarse error is at roundoff level, which avoids meaningless rates for analytically zero components. The production pass/fail checks compare observed rates with expected rates rather than testing only monotone error decrease.

18 Output Diagnostics

Convergence CSV files include DG order, ESPRK order, mesh family, mesh level, element count, owned-element distribution, mesh sizes, mesh quality, aggregate errors and rates, per-component errors and rates, available invariant diagnostics, and runtime. Production transient drivers additionally write time-series ParaView output, quadrature diagnostics, run metadata, partition metadata, integration-point samples, and checkpoints.